

Parabolik KDD için Geri Fark Metodu.

$$\frac{\partial u}{\partial t} - \alpha^2 \frac{\partial^2 u}{\partial x^2} = 0 \quad 0 \leq x \leq \ell, t > 0$$

$$u(0, t) = g t_0(0, t)$$

$$u(L, t) = g t_L(L, t)$$

$$u(x, 0) = g x_0(x, 0)$$

$$h = \ell/m$$

k : t için adım uzunluğu.

	$x_1 = 0$	$x_2 = h$	$x_3 = 2h$	...	$x_m = (m-1)h$	$x_{m+1} = mh = \ell$
$t_1 = 0$	$w_{1,1}$	$w_{2,1}$	$w_{3,1}$	...	$w_{m,1}$	$w_{m+1,1}$
$t_2 = k$	$w_{1,2}$	$w_{2,2}$	$w_{3,2}$	...	$w_{m,2}$	$w_{m+1,2}$
$t_3 = 2k$	$w_{1,3}$	$w_{2,3}$	$w_{3,3}$	...	$w_{m,3}$	$w_{m+1,3}$
$\vdots$						

$$j = 1, 2, 3, \dots$$

$$w_{1,j} = g t_0(0, t_j)$$

$$w_{m+1,j} = g t_L(L, t_j)$$

$$i = 1, 2, \dots, m+1$$

$$w_{i,1} = g x_0(x_i, 0)$$

Sonlu Fark Formülleri

$$\frac{\partial u}{\partial t}(x_i, t_j) \approx \frac{w_{i,j} - w_{i,j-1}}{k}$$

$$\frac{\partial^2 u}{\partial x^2}(x_i, t_j) \approx \frac{w_{i+1,j} - 2w_{i,j} + w_{i-1,j}}{h^2}$$

$$\lambda = \frac{\alpha^2 k}{h^2} \quad \text{ve} \quad 1 + 2\lambda = M \quad \text{olmak üzere}$$

$$M w_{i,j} - \lambda w_{i+1,j} - \lambda w_{i-1,j} = w_{i,j-1}$$

geri sonlu
fark
denklemini

Son sıcaklık değeri T olsun. 0 zaman $k = T/n \rightarrow$ alt aralık
sayısı.

$n+1$ tane j değeri var.

$m+1$ tane de i " " .

$j=2$ için

$$i=2 \quad M w_{2,2} - \lambda w_{3,2} = w_{2,1} + \lambda w_{1,2}$$

$$i=3, \dots, m-1 \quad M w_{i,2} - \lambda w_{i+1,2} - \lambda w_{i-1,2} = w_{i,1}$$

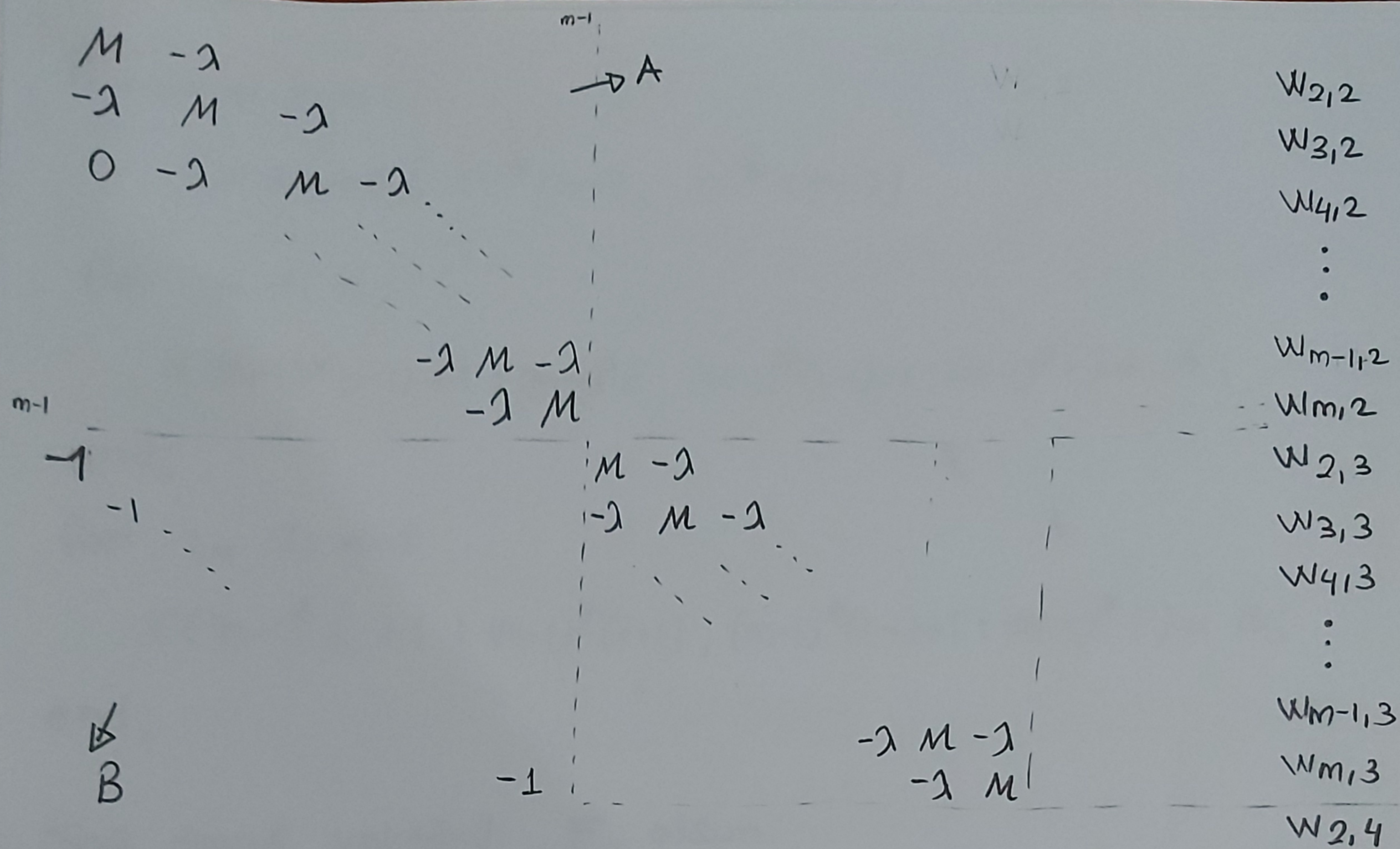
$$i=m \quad M w_{m,2} - \lambda w_{m-1,2} = w_{m,1} + \lambda w_{m+1,2}$$

$j=3, \dots, n+1$ için

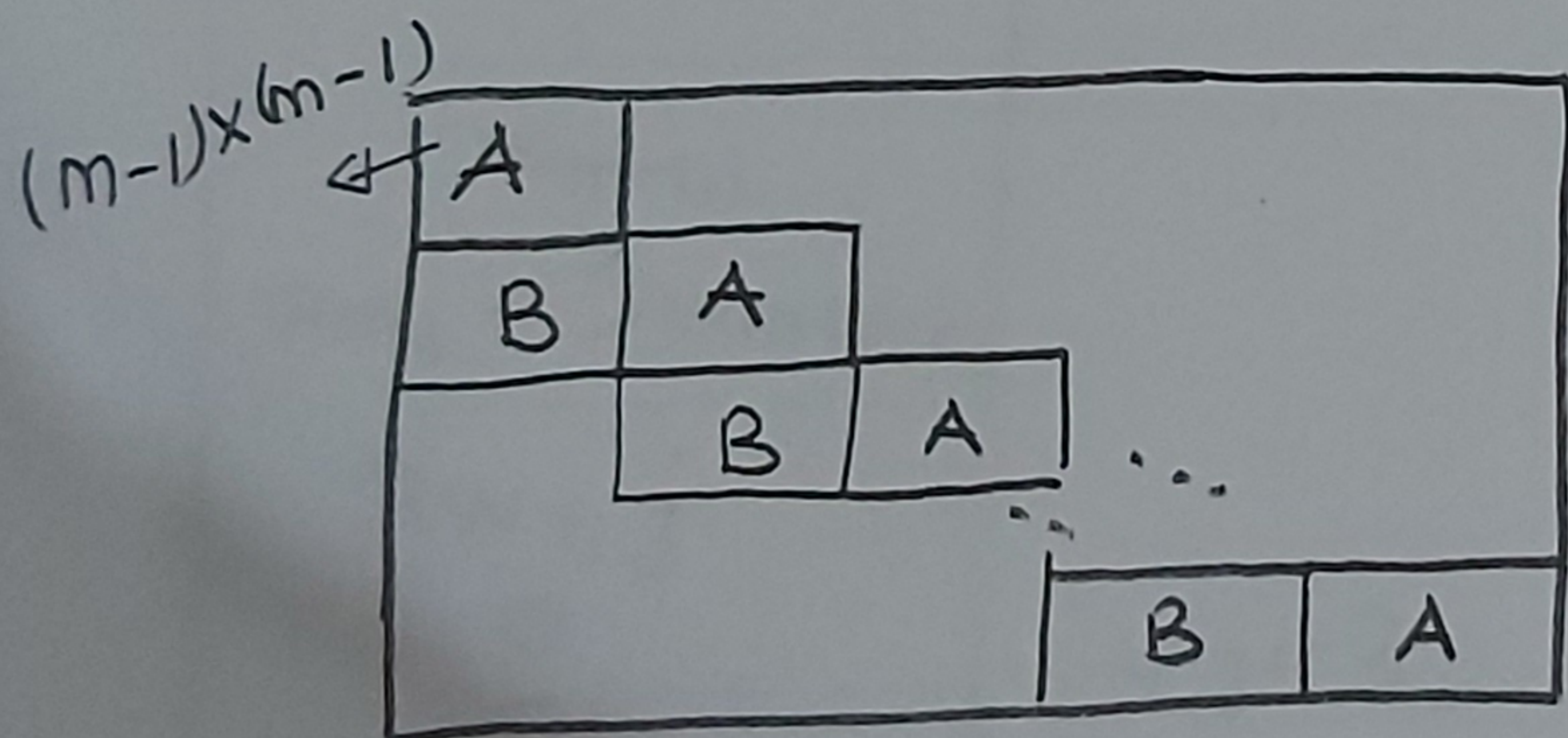
$$i=2 \quad M w_{2,j} - \lambda w_{3,j} - w_{2,j-1} = \lambda w_{1,j}$$

$$i=3, \dots, m-1 \quad M w_{i,j} - \lambda w_{i+1,j} - \lambda w_{i-1,j} - w_{i,j-1} = 0$$

$$i=m \quad M w_{m,j} - \lambda w_{m-1,j} - w_{m,j-1} = \lambda w_{m+1,j}$$



Katsayılar matrisi K



$A = \text{zeros}(m-1, m-1)$

for $i = 1 : m-2$

$A(i, i+1) = -\lambda;$

$A(i+1, i) = -\lambda;$

$A(i, i) = M;$

end

$A(m-1, m-1) = M;$

$B = \text{zeros}(m-1, m-1)$

for $i = 1 : m-1$

$B(i, i) = -1;$

end;

K matrisi için

$$K = \text{zeros}((n * (m-1)), (n * (m-1)))$$

for i = 1: n

$$K((m-1)*(i-1)+1 : (m-1)*i, (m-1)*(i-1)+1 : (m-1)*i) = A;$$

end;

for i = 1: n-1

$$K((m-1)*i+1 : (m-1)*(i+1), (m-1)*(i-1)+1 : (m-1)*i) = B;$$

end;

Sağ taraf vektörü F olsun.

$$F = \begin{bmatrix} w_{2,1} + \lambda w_{1,2} \\ w_{3,1} \\ w_{4,1} \\ \vdots \\ w_{m-1,1} \\ w_{m,1} + \lambda w_{m+1,2} \\ \hline \lambda w_{1,j} \\ 0 \\ 0 \\ \vdots \\ 0 \\ \lambda w_{m+1,j} \end{bmatrix}$$

$\rightarrow j = 3, \dots, n+1$

$$F = \text{zeros}((n-1)*(m-1), 1)$$

$$F(1,1) = g x_0(x_2, 0) + \lambda^* g t_0(0, t_2)$$

$$F(m-1,1) = g x_0(x_m, 0) + \lambda^* g t_L(0, t_2)$$

for i = 2: m-2

$$F(i,1) = g x_0(x_{i+1}, 0);$$

end;

for j = 3: n+1

$$F((m-1)*(j-2)+1,1) = \lambda^* g t_0(0, t_j)$$

$$F((m-1)*(j-1),1) = \lambda^* g t_L(0, t_j)$$

end;

$$W = K \setminus F;$$

Parabolik KDD için Crank Nicolson Metodu.

Sonlu Fark Formülü

$$\frac{w_{i,j+1} - w_{i,j}}{k} - \frac{\alpha^2}{2} \left(\frac{w_{i+1,j} - 2w_{i,j} + w_{i-1,j}}{h^2} + \frac{w_{i+1,j+1} - 2w_{i,j+1} + w_{i-1,j+1}}{h^2} \right) = 0$$

$$- \frac{1}{2} \frac{k\alpha^2}{h^2} w_{i+1,j+1} + \left(1 + \frac{\alpha^2 k}{h^2} \right) w_{i,j+1} - \frac{1}{2} \frac{k\alpha^2}{h^2} w_{i-1,j+1}$$

$$= \frac{1}{2} \frac{k\alpha^2}{h^2} w_{i+1,j} + \left(1 - \frac{\alpha^2 k}{h^2} \right) w_{i,j} + \frac{1}{2} \frac{k\alpha^2}{h^2} w_{i-1,j}$$

$$\lambda = \frac{k\alpha^2}{h^2}, \quad 1 + \lambda = M1, \quad 1 - \lambda = M2 \quad \text{olmak üzere}$$

$$- \frac{\lambda}{2} w_{i+1,j+1} + M1 \cdot w_{i,j+1} - \frac{\lambda}{2} w_{i-1,j+1} = \frac{\lambda}{2} w_{i+1,j} + M2 \cdot w_{i,j} + \frac{\lambda}{2} w_{i-1,j}$$

Crank Nicolson
metodu formülü.

$j=1$ için

$$i=2 \text{ için } -\frac{\lambda}{2} w_{3,2} + M1 \cdot w_{2,2} = \frac{\lambda}{2} w_{3,1} + M2 w_{2,1} + \frac{\lambda}{2} w_{1,1} + \frac{\lambda}{2} w_{1,2}$$

$$i=3 \dots m-1 \text{ için } -\frac{\lambda}{2} w_{i+1,2} + M1 \cdot w_{i,2} - \frac{\lambda}{2} w_{i-1,2} = \frac{\lambda}{2} w_{i+1,1} + M2 w_{i,1} + \frac{\lambda}{2} w_{i-1,1}$$

$$i=m \text{ için } M1 \cdot w_{m,2} - \frac{\lambda}{2} w_{m-1,2} = \frac{\lambda}{2} w_{m+1,1} + M2 w_{m,1} + \frac{\lambda}{2} w_{m-1,1} + \frac{\lambda}{2} w_{m+1,2}$$

Önce bu denklem sistemi çözülür. Yani $w_{i,2}$ 'ler hesaplanır. Sonra aynı biçimdeki $w_{i,3}$ 'ler bilinen $w_{i,2}$ 'ler kullanılarak hesaplanır. Bu işlem $j=n$ 'e kadar sürdürülür.

$$\begin{bmatrix} M1 & -\lambda/2 & & & \\ -\lambda/2 & M1 & -\lambda/2 & & \\ & -\lambda/2 & M1 & -\lambda/2 & \\ & & & \ddots & \\ & & & -\lambda/2 & M1 & -\lambda/2 \\ & & & & -\lambda/2 & M1 \end{bmatrix} \begin{bmatrix} w_{2,2} \\ w_{3,2} \\ w_{4,2} \\ \vdots \\ w_{m-1,2} \\ w_{m,2} \end{bmatrix}$$

$$= F = \begin{bmatrix} \lambda/2 w_{3,1} + M2 w_{2,1} + \lambda/2 w_{1,1} + \lambda/2 w_{1,2} \\ \lambda/2 w_{4,1} + M2 w_{3,1} + \lambda/2 w_{2,1} \\ \vdots \\ \lambda/2 w_{m,1} + M2 w_{m-1,1} + \lambda/2 w_{m-2,1} \\ \lambda/2 w_{m+1,1} + M2 w_{m,1} + \lambda/2 w_{m-1,1} + \lambda/2 w_{m+1,2} \end{bmatrix}$$

$$j=2 \text{ i4in}$$

$$i=2 \text{ i4in} \quad -\frac{\lambda}{2} w_{3,3} + M1 \cdot w_{2,3} = \frac{\lambda}{2} w_{3,2} + M2 \cdot w_{2,2} + \frac{\lambda}{2} w_{1,2} \\ + \frac{\lambda}{2} w_{1,3}$$

$$i=3 \dots m-1 \text{ i4in} \quad -\frac{\lambda}{2} w_{i+1,3} + M1 w_{i,3} - \frac{\lambda}{2} w_{i-1,3} \\ = \frac{\lambda}{2} w_{i+1,2} + M2 w_{i,2} + \frac{\lambda}{2} w_{i-1,2}$$

$$i=m \text{ i4in} \quad M1 \cdot w_{m,3} - \frac{\lambda}{2} w_{m-1,3} = \frac{\lambda}{2} w_{m+1,2} + M2 \cdot w_{m,2} \\ + \frac{\lambda}{2} w_{m-1,2} + \frac{\lambda}{2} w_{m+1,3}$$